Natural Language Processing

Lab Week 6: Inference with LLMs with the HuggingFace Transformers Library

2025/26

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1. The HuggingFace transformers library is a 'definition framework' for research models in AI, and also contains some implementation of popular widely-used models along with examples and tutorials. You will need to make an account with HuggingFace, and you may have to choose a smaller model than the default settings in some cases if running the examples on the lab machines.
   (a) Follow the installation process here: `https://huggingface.co/docs/transformers/en/installation`. (note that you only need to do one of the options - pip, uv, conda, or source install)
   (b) Follow the quickstart guide. Set up a HuggingFace account and see how it works with your hardware. `https://huggingface.co/docs/transformers/en/quicktour`
   (c) Follow their 'prompt engineering' tutorial: `https://huggingface.co/docs/transformers/en/tasks/prompting`. Experiment with editing the example code they provide.
   (d) Follow their 'zero-shot learning' tutorial: `https://huggingface.co/docs/setfit/en/tutorials/zero_shot`. The example demonstrates sentiment analysis.
   (e) On the Moodle there are three more example scripts showing natural language inference, instruction tuning, and summarisation. For natural language inference, this model on HuggingFace shows some good examples: `https://huggingface.co/tasksource/deberta-small-long-nli`